Retrieval-Augmented Generation Explained

Retrieval-augmented generation, or RAG, is a technique that combines a search step with a language model generation step. Instead of relying only on what a model learned during training, RAG retrieves relevant text from an external knowledge source at query time and includes it in the prompt.

The retrieval step typically works by embedding documents into vectors using a model such as sentence-transformers, storing those vectors in a database like Chroma or FAISS, and then finding the nearest vectors to a query at question time.

Chunk size is an important design decision. If chunks are too large, retrieval returns noisy context that dilutes the relevant information. If chunks are too small, individual chunks lose surrounding context and meaning.

A common failure mode is hallucination, where the model generates a confident-sounding answer that is not actually supported by the retrieved context. Requiring citations back to source pages is one way to reduce this risk and make failures easier to detect.